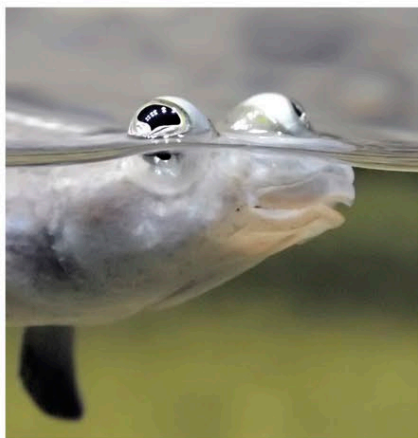


## Vision

Eyes packed with light-sensitive cells enable animals to see. Vertebrates, such as humans, have two camera-like eyes that focus light onto the back of the eye. But some invertebrates rely on many more eyes—the giant clam has hundreds of tiny eyes scattered over its body. Each animal's eyes are specialized in different ways. Some are so sensitive that they can pick up the faintest light in the dark of night or in the deep sea.



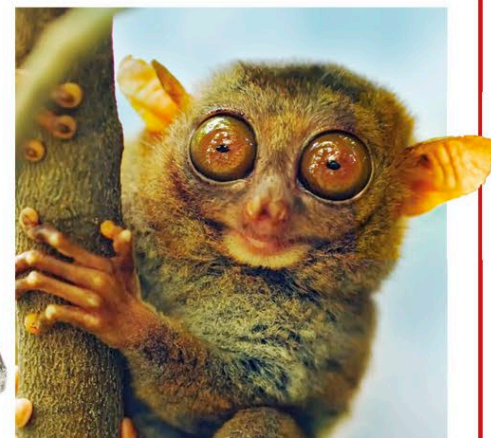
### Four-eyed fish

When it swims at the surface, this fish's split-level eyes help it to focus on objects above and below the water.



### Long-legged fly

Flies and many other insects have compound eyes—made up of thousands of tiny lenses.



### Tarsier

This primate's eyes are the biggest of any mammal when compared to the size of its head. They help it see well at night.

## Touch

Animals have receptors in their skin that sense when other things come into contact with their body. Some receptors only pick up firm pressures, while others are sensitive to the lightest of touches. Receptor cells are especially concentrated in parts of the body that rely a lot on feeling textures or movement. Human fingertips are crammed with touch receptors, as are the whiskers of many cats, and the unusual nose of the star-nosed mole.



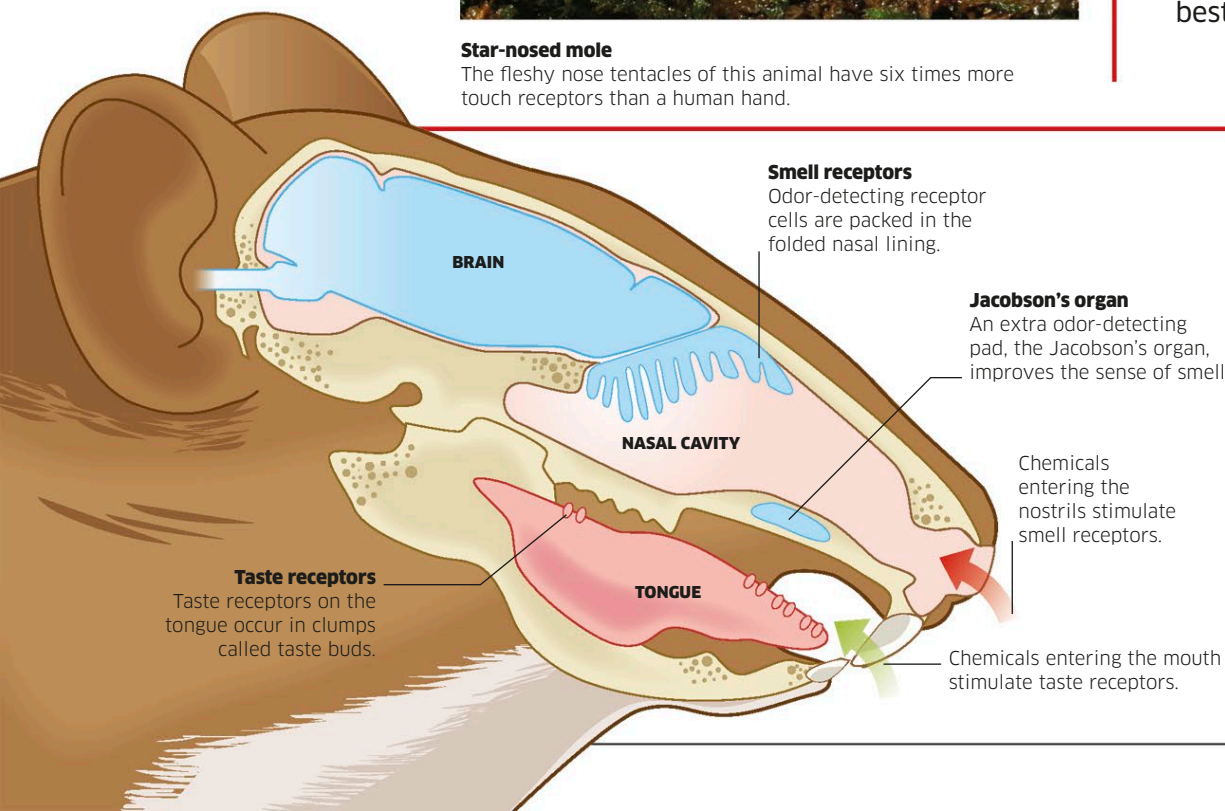
### Star-nosed mole

The fleshy nose tentacles of this animal have six times more touch receptors than a human hand.

# Senses

**Animals sense their surroundings using organs that are triggered by light, sound, chemicals, or a whole range of other cues.**

Sense organs are part of an animal's nervous system. They contain special cells called receptors that are stimulated by changes in the environment and pass on signals to the brain and the rest of the body. Through these organs animals can gain a wealth of information about their surroundings, equipping them to react to threats or opportunities. Each kind of animal has sense organs that are best suited to the way it lives.



## Taste and smell

Smelling and tasting are two very similar senses, as they both detect chemicals. The tongue has receptors that taste the chemicals dissolved in food and drinks, and receptors inside the nose cavities pick up the chemicals in odors. Some animals that are especially reliant on chemical senses, or that do not have receptors elsewhere, have a concentrated patch of receptors in the roof of their mouth, called a Jacobson's organ.

### Mouse senses

Like most mammals, a mouse has a keen nose. It uses smell to communicate with others of its kind: signaling a territorial claim or a willingness to mate. A mouse's tongue detects tastes in food, and both tongue and nose send signals to the brain.